

Scalability & Future Plan

Date	19 July 2024
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single SQLite database used for storing applicant records	May slow down as applicant records grow	High
2	No user authentication for analysts	Limits access control for the platform	Medium
3	Model trained on a static historical dataset	May not reflect newer applicant trends	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles a few users on a local Flask server	Deploy on IBM Cloud or Render with load balancing for concurrent users
2	Data Storage	SQLite database for applicant records	Migrate to a scalable cloud database such as PostgreSQL
3	Performance	Single model inference per request	Cache frequent predictions and optimize model loading
4	Security	No login or access control	Add authentication and encrypt applicant data in transit

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
-------	-------------------------------	-----------------	-----------------

Phase 2	Add user authentication and role-based access	Next 2 months	Improved security and controlled access
Phase 3	Migrate to a scalable cloud database	Next 4 months	Supports larger applicant data volumes
Phase 4	Add batch prediction and analytics dashboard	Next 6 months	Better insights for compliance officers